Introduction

Gout is a heterogeneous disease caused by deposition of monosodium urate crystals, characterized as hyperuricemia (HUA), recurrent acute arthritis, gouty tophus sedimentation, chronic arthritis and joint deformity [1]. Many evidences showed that hyperuricemia and gout are independently risk factors for chronic kidney disease, hypertension, cardiovascular and cerebrovascular diseases and diabetes [2]. Recently, due to the change of living condition and peoples' food structure, especially with the increasing consumption of high-purine food, such as seafood, broth and animal viscera, the prevalence of hyperuricemia and gout is significantly increasing [3–5]. Several European and North American studies reported gout prevalence in the range of 1.9 and 3.9% [4], and prevalence of HUA and gout in China was 13.3% and 1.1%, respectively [6].

It is commonly accepted that two-thirds of the urate is excreted from the kidney into urine via the 'renal excretion' pathway, and the remaining one-third via the 'extra-renal excretion' pathway [7]. Taking extra-renal urate excretion into account, the current classification of hyperuricemia and gout is based on patients' urinary urate excretion (UUE) and fractional excretion of urate clearance (urate clearance/creatinine clearance ratio, FE_{UA}). Previous studies have shown that the type of hyperuricemia classification might affect the curative effect of urate-lowering therapy [8–10]. Our latest study also indicated for patients of renal uric acid underexcretion type, benzbromarone had superior urate-lowering compared to Febuxostat [11].

Gout is related to heredity, environment, race, dietary and lifestyle factors. Long-established dietary risk factors, including meat, seafood, alcohol, sweet food, fructose beverage, is known to play an important role in HUA and gout [1, 12–15]. Diet adjustment has become a supplement to drug treatment of gout patients. It was reported that simple dietary regimens may be beneficial to gout patients [16]. A cohort study convincingly demonstrated that vegetarians, who avoiding purine-rich meat and seafood, while consuming more vegetables, whole grains, seeds and nuts [17, 18], experienced a lower risk of gout (without adjustment for hyperuricemia: HR: 0.33; 95%*CI* 0.14, 0.79; with adjustment for hyperuricemia: HR: 0.40; 95%*CI* 0.17, 0.97) [19]. Previous investigations indicate that sUA level of HUA patients decreased by 60–120 μmol/L on average after the intervention of purine-free diet for 7–10 days [20]. So far, it was unclear that effect of low-purine loading test on sUA of gout patients based on different clinical types. Thus, the present study aims to evaluate the effect of low-purine diet on sUA and other biochemical indexes levels of gout patients in different clinical types and provide evidence-based solutions for doctors and patients.

Materials and methods

Participants

Adult primary gout patients were recruited from 2019 to 2022. The diagnostic criteria of gout refer to the classification criteria of gout jointly developed by ACR/EULAR in 2015 [3]. Any patients with severe malnutrition, tumors, serious kidney, liver, or heart diseases, history of standardized urate-lowering therapy (ULT), or an acute gout attack within two weeks were excluded. The patients were divided into three clinical types according to the FE_{UA} and UUE: (1) urate 'overproduction' type: UUE > 600 mg/d/1.73 m^2 and $FE_{UA} \geq 5.5\%$; (2) 'underexcretion' type: UUE ≤ 600 mg/d/1.73 m^2 and $FE_{UA} < 5.5\%$; (3) 'combined' type: UUE > 600 mg/d/1.73 m^2 and $FE_{UA} < 5.5\%$ [3, 20, 21]. Written informed consent was obtained from all participants.

Study design

The single-center prospective cohort study was conducted in Shandong Gout Clinic Center in the Affiliated Hospital of Qingdao University. The study was approved by the Ethics Committee of the Affiliated Hospital of Qingdao University (QYFY WZLL 26607) and registered at the China Clinical Trial Registration Center (No. ChiCTR1900022981) at 06/05/2019.

Before the study participants attended group and individual meetings for a detailed explanation of the study design and diet requirements. Each patient was given a low-purine diet tailored by a nutritionist. The low-purine diet in this experiment aims to limit the purine content in food to below 200 mg per day. The recommended diet is low in fat (up to 30% of total caloric value), high in carbohydrates, low in protein (0.8 g/kg/day), and limited in the consumption of foods high in purines (100–1000 mg/100 g). These foods include viscera, consommé, meat extract, fish (like herring, mackerel, and sardines), mussels, anchovies, alcoholic beverages, yeast, foods containing eggs and yeast, and partridge. It is also advisable to limit consumption of foods with moderate purine content (9 to 100 mg per 100 g), including meats, seafood, and certain vegetables. Additionally, it is recommended to avoid foods high in oxalates, such as dark green vegetables, cauliflower, beet/chard, beetroot, eggplant, okra, sweet potatoes, chestnuts, coconuts, wheat germ, tomatoes, asparagus, mushrooms, strawberries, whole wheat cereals, quince marmalade, and chocolate. The recommended daily liquid intake is 2.5 to 3.5 L [22]. All patients were expected to under low-purine diet for 2 weeks and underwent a follow-up visit to evaluate the effects of the low-purine loading on biochemical tests. The compliance

to the diet intervention was assessed during the follow-up visit by asking the patients and questionnaire. Patients were also advised to do regular moderate-intensity exercise (≥30 min a day) for both normal-weight and overweight. Patients were required not to take urate-lowering drugs and/or drugs affecting serum urate for 2 weeks. It should be specifically noted that although the population of the current study overlaps with that of other cohort studies conducted by the same group, there was no noninterference between interventions because other research interventions, such as oral urate-lowering drugs, were conducted carried out after the completion of the current study.

Data collection

The general information including gender, age, occupation, smoking and drinking habits, onset age, attack frequency, comorbidity, family history of gout, medication history were collected at the baseline visit (PRE).

Before and after LPD, height, weight, systolic blood pressure (SBP) and diastolic blood pressure (DBP) were measured with unified tools and standard measurement methods. The biochemical parameters, such as sUA, ALT, AST, FPG, TG, CH, BUN, Scr, creatinine clearance rate (Ccr), urine creation (Cr-U) and urine uric acid (UA-U) were collected by automatic biochemical analyzer (Roche Cobas C501). Body mass index (BMI) was calculated as weight (kg) divided by height (m) squared. The estimating glomerular filtration rate (eGFR) was calculated followed the modified MDRD equation. The creatinine clearance rate (Ccr) were used to assess the renal function [Cockcroft–Gault (CG) equation, Ccr = (140-age) × body weight (kg)/0.818 × Scr (μmol/L)]. The FEUA (uU/uCr × sCr/sUA × 100%) and UUE (uU × 24 h urinary volume/(0.0061 × height (cm) + 0.0128 × weight (kg) − 0.1529) × 1.73 (mg/d/1.73 m^2)) were tested using 24-h urine samples.

Outcomes

The main outcomes were the change of sUA level in subgroups after 2 weeks low-purine diet. The physical and chemical indicators (SBP, DBP, BMI, TG, TC, liver and kidney function, etc.) were the secondary endpoints. Associations of change of sUA and clinical variables were analyzed.

Statistical analysis

Statistical analysis was performed using SPSS 25.0 software (SPSS, Inc., Chicago, IL, USA). Data were tested for normal distribution using the Shapiro–Wilk test and for equality of variances when needed (Levene's test). For continuous variables, the results were described as the mean ± standard or median (25–75th percentile), while the categorical variables were described as percentages (%). A one-way analysis of variance (ANOVA) or Kruskal–Wallis ANOVA was used for continuous variables and a Chi-square test was used to assess the distribution of categorical variables. Differences of measurement data inside group were compared with paired samples *t*-test or Wilcoxon rank-sum test, a one-way analysis of variance (ANOVA) or Kruskal–Wallis ANOVA was used to compare among groups. The baseline sUA level was corrected by covariance analysis to obtain the uric acid level after the test. Multiple stepwise linear regression was used to analyze the independent risk factors of uric acid change (end-of-period minus baseline) between baseline and follow-up with BMI, family history, average weekly exercise, daily water quantity, tophi, sUA and other biochemical indexes levels at baseline being independent variables and variation of sUA level (ΔsUA) being dependent variables, respectively. A p value < 0.05 was defined as statistically significant.

Results

Baseline population characteristics

During the low-purine diet, 43 patients failed to complete the course and 26 patients were lost to follow-up. Ultimately, 626 participants (69 in the overproduction type, 428 in underexcretion type and 129 in the combined type) completed the study and were included in the outcome analyses (Fig. 1).

Table 1 shows clinical characteristics of the total gout patients. Mean age and BMI were 41.20 (± 13.41) year and 27.53 (± 3.63) kg/m^2. Most of the total 626 subjects were males (98%). 41.5% of the patients with gout reported a history of smoking, 65.8% of them reported a history of drinking alcohol and 17.6% had tophi. The combined type has been characterized by a higher baseline BMI, sUA and eGFR than other two types. We also observed younger age in combined type than overproduction type but did not differ from underexcretion type. TC was observed to be highest in overproduction type, which was significantly higher than underexcretion type, but did not reach statistical significance from combined type.

Comparison of outcome measures and clinical characteristics in total population

Overall, SBP, DBP, BMI, ALT, AST, TG, TC, Scr, BUN and sUA of 626 gout patients were decreased significantly after 2-week low-purine diet. Serum urate dropped from 576.43 ± 91.29 μmol/L to 514.01 ± 88.17 μmol/L (Table 2).

Comparison of primary outcomes in subtypes

After two weeks of low-purine diet, we observed that sUA were significantly decreased in overproduction